

PROBLEM 2 (1 hour)

An experimental airplane is powered by an engine without moving parts that delivers a heat $\dot{Q}$ per unit time. The entrance and exit sections of the engine are, respectively, below and above the airplane's center of gravity (CoG), which is placed at the origin of the coordinate system shown in the figure. Both sections are rectangular with the same spanwise length b (along the z -direction), and heights a_i and a_o , as indicated in the figure. The flow at the engine's inlet has velocity V_i , pressure p_i and density ρ_i , while the outlet jet discharges at pressure p_o .

- The flow is steady.
 - Gravitational forces are negligible.
 - The conditions at the engine's inlet and outlet sections are uniform.
 - The air behaves as a perfect gas of constant R_g and specific heat ratio γ .
 - The known data are $(a_i, V_i, \rho_i, p_i, a_o, p_o, p_a, \dot{Q})$.
1. Applying the mass and energy conservation equations to an appropriately defined control volume, obtain the pair of equations that determine the jet exhaust velocity, V_o , and the corresponding density, ρ_o . In the remainder of the problem, assume that V_o and ρ_o are already known. **(4 points)**
 2. Use the momentum equation to obtain the thrust provided by the engine, $\bar{T}$. **(2 points)**
 3. Use the angular momentum equation to obtain the net torque $\bar{M}$ acting on the airplane with respect to its CoG. Assume that the surrounding air is at uniform pressure p_a , except at the engine's inlet and outlet. **(3 points)**
 4. To compensate the torque obtained in the previous question, the engineers decide that the main wing's CoG coincides with that of the full airplane, so that it doesn't contribute to the torque, and to place a control surface (canard) at a distance L from the CoG (see figure). Knowing that this control surface provides a lift force F , determine the distance L . **(1 point)**

